# Introduction

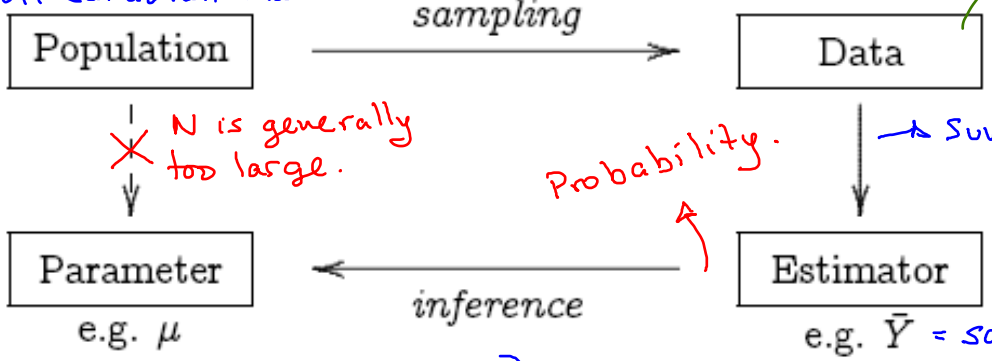
**Statistical Inference:** Drawing conclusions about ***populations*** (**parameters**) based on information collected in ***samples*** (**statistics**).

Statistical inference goes as follows:

e.g. Adult Canadian Males

incomplete information.

$n$ = sample size.
$Y_1, Y_2, \ldots, Y_n$




N is generally too large.

Summarize, organize, graph.

Probability.

or Statistic

e.g. $\bar{Y}$ = sample mean = $\dfrac{Y_1 + Y_2 + \cdots + Y_n}{n}$.

e.g. $Y$ = weight.

$\mu_Y$ = population mean weight.

$N$ = population size (millions ---)

$$\mu_Y = \frac{Y_1 + Y_2 + \cdots + Y_N}{N}.$$

What does $\bar{Y}$ imply about $\mu_Y$?

<u>Note</u>: Prior to sampling $\bar{Y}$ is a random variable.

**Illustration :**

Suppose our goal is to answer a question about some population mean $\mu$.

To get started we can collect some information in the form of a sample and estimate the paramater. Let's consider the sample mean as our estimate.

If $Y_1, Y_2, ..., Y_n$ represents our random sample from this population, where $Y_1, Y_2, ..., Y_n$ are identically distributed with $E[Y_i] = \mu,\ i = 1, ..., n,$ the sample mean is $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (a function of the random variables in the sample).

**Parameter :** $\mu$ - the (unknown) population mean

**Estimator :** $\bar{Y}$ - the sample mean

The key to inference is understanding the properties of the estimator $\bar{Y}$.

e.g. Suppose we observed $\bar{y} = 175$ pounds from a sample of $n = 20$.

Questions to consider:

What does the sample imply about the population? Or, what does the estimate imply about the paramater.

- Is $\bar{Y}$ equal to $\mu$? Is $\mu_Y = 175$? ⇒ NO!
- Is $\bar{Y}$ close to $\mu$? Hopefully...
- How close can we $\underbrace{\text{expect } \bar{Y} \text{ to be to } \mu}_{\text{Sampling variability. Error in estimation.}}$? With what certainty? → or confidence. ⇒ Based the prob. dist. of $\bar{Y}$.

Beyond the questions above, we would also like to know why we are using $\bar{Y}$ to estimate $\mu$? Is that a good choice? The best choice? Are there other estimates we should consider? ↳ Not always.

All of these questions will be answered in due time.

# Unit 6: Distributions of Functions of Random Variables

Estimates are functions of random variables. In order to understand the properties of our estimates, we need to understand their probability distribution.

The goal of Unit 6 is to explore different methods to help identify the probability distributions of functions of random variables.

- ⇨ Three Methods
- ⇨ Order Statistics : what's the distribution of
  - minimum( $Y_1, Y_2, \ldots, Y_n$ ) ?

# Unit 7: Sampling Distributions

The *Normal* probability model plays a special role in inferential statistics.

The goal of Unit 7 is to define the probability distribution for functions of random variables collected from *Normally* distributed populations.

- ⇨ Distributions related to the Normal Model.
- ✳ ⇨ Central Limit Theorem

If $Y_1, Y_2, \ldots, Y_n$ is a sample from a Normal$(\mu_Y, \sigma_Y^2)$ distribution, what is the distribution of

- $\bar{Y}$?
- $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2 \Rightarrow$ sample variance?

# Unit 8: Point Estimation

In statistical inference we need to estimate the population parameters based on sample statistics. Any function of observations collected in a sample is a statistic. Thus, there are an infinite number of statistics we could use.

The goal of Unit 8 is to explore ideal properties of a statistic and methods to help identify which statistic(s) we should use.
and to find the MVUE = minimum variance unbiased estimate.

- ⇨ Point Estimation. Properties of Point Estimates
- ⇨ Methods of Estimation

e.g. Suppose $Y \sim \text{Gamma}(\underbrace{\alpha, \beta}_{\text{Parameter.}})$. What is an estimate for $\alpha$? $\beta$?

Suppose $Y \sim \text{Uniform}(\theta_1, \theta_2)$. What is an estimate for $\theta_1$? $\theta_2$?

# Unit 9: Interval Estimation (Confidence Intervals)

Point estimates are single values calculated to estimate a parameter. These point estimates contain error (another sample will result in different observations). We want to account for this error (sampling variability). We will use this information to estimate our parameters with an interval of values. This interval will be reported with a chosen level of confidence. Thus, the interval will be called a confidence interval.

The goal of Unit 9 is to find an interval of values of which we have (hopefully) a high level of confidence that the parameter lies.

⇨ Interval Estimation (Confidence Intervals)

e.g. Suppose we observed $\bar{y} = 175$ pounds from $n = 20$.

- Point Estimation: We estimate $\mu_Y$ to be 175 pounds.
- Confidence Estimation: We estimate $\mu_Y$ to be $175 \pm 8.5$, or between 166.5 pounds and 183.5 pounds, with 98% confidence.

# Unit 10: Hypothesis Testing

Sometimes there is a question about a specific value for a parameter. e.g. Is $\mu_Y > 170$ pounds?
Does $\bar{y} = 175$ from $n = 20$ imply that $\mu_Y > 170$?

The goal of Unit 10 is to explore how to test the validity of the value in question.

- ⇨ Elements of a test.
- ⇨ Type I and II errors.
- ⇨ Significance level, power and p-value.
- ⇨ Likelihood Ratio Tests

# Unit 6: Distributions of Functions of Random Variables

The goal of Unit 6 is to explore different methods to help identify the probability distributions of functions of random variables.

Consider $U = h(Y_1, \ldots, Y_n)$ to be a function of the random variables $Y_1, \ldots, Y_n$, where the joint distribution of $Y_1, \ldots, Y_n$ is known.

What is the probability distribution of $U$?

- ⇨ Three Methods
- ⇨ Order Statistics

## 6.1: Introduction (done)

## 6.2: A summary of the methods to be discussed

**Three Methods:**

**6.3: Method of Distribution Functions**

**6.4, 6.6: Method of Transformations**

**6.5: Method of Moment-Generating-Functions**

## 6.7: Order Statistics